

ASSIGNMENT-04

AIM :- Enhance web page developed in earlier assignment by rendering Lists & Portals, Error Handling, Routers & style with React CSS also make it a responsive design to the scale well across PC, tablet & Mobile Phone.

OBJECTIVES :-

- 1) Enhance UI & Experience.
- 2) Improve Application Robustness & Navigation.

THEORY :-

Q1) How do List & keys work in React?

Ans- Lists in React are usually created using the `.map()` function to render multiple components dynamically.

Keys are unique identifiers assigned to each list item to help React differentiate elements during updates.

They improve rendering efficiency by allowing React to re-render only changed elements, not the entire list.

A good practice is to use stable, unique IDs as keys to avoid rendering issues.

Q2) What is a React Portal & when would you use one?

Ans - A portal allows you to render a component's children into a DOM node that exists outside the parent component's hierarchy.

Syntax :- ReactDOM.createPortal(child, container)

Useful for :-

- Avoiding CSS overflow/positioning issues.

- Modals, dialogs, tooltips, dropdowns.

Q3) Discuss the importance of Error Boundaries in React.

Ans - Error Boundaries are React components that catch JavaScript errors in their child component tree.

- They prevent the entire app from crashing when an error occurs.

- They can show a fallback UI instead of breaking the whole app.

- Crucial for production apps where stability & user experience matter.

Q4) How does React Router enable Single Page Application (SPA) functionality?

Ans - React Router manages navigation without reloading the page.

- It uses the History API to update the URL while keeping the app alive.

- Components are rendered conditionally based on the route, enabling seamless transitions.
- Supports nested routes, dynamic parameters, redirects & lazy loading for efficiency.

Q5) Explain the different ways to style a React application.

✓ Inline Styling :- Using the style attribute with objects.

CSS Stylesheets :- Importing regular .css files & applying class names.

CSS Modules :- Scoped styles using .module.css files to avoid naming conflicts.

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